

1. Administrative & Study Identification

Full Study Title: A Phase IIIB Multicenter, Randomized, Double-blind, Controlled Study to Evaluate the Efficacy, Safety and Pharmacokinetics of a Higher Dose of Ocrelizumab in Adults With Relapsing Multiple Sclerosis **Short Title/Acronym:** MUNETTE Study **Posting ID:** 909857978771445983 **Protocol Number:** BN42082 (EudraCT: 2020-000893-69) **NCT Number:** NCT04544436 **Sponsor Name:** Roche **Investigational Product:** Ocrelizumab (higher dose vs. approved dose 600 mg) **Phase of Development:** Phase 3 **Trial Status:** Active, not recruiting **Medical Monitor:** Roche Clinical Operations Lead / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether a higher dose of ocrelizumab (1200 mg or 1800 mg based on baseline body weight) significantly delays the time to onset of 12-week composite confirmed disability progression (cCDP12) compared to the currently approved 600 mg dose in participants with Relapsing Multiple Sclerosis (RMS). **Secondary Objectives:** To assess the efficacy, safety, and pharmacokinetics (PK) of higher doses of ocrelizumab, evaluating time to onset of 24-week cCDP (cCDP24), 48-week cCDP (cCDP48), cCDP12 independent of protocol-defined relapses (PDR), and the incidence of adverse events.

2.2 Background & Rationale To address the ongoing risk of disability progression in patients with RMS. While ocrelizumab at the approved 600 mg dose is highly effective at reducing relapses and MRI activity, pharmacokinetic-pharmacodynamic modeling and observational data suggested that higher ocrelizumab exposure could further reduce the risk of disability progression. This study was initiated to determine if a higher-than-approved dose of ocrelizumab provides superior clinical outcomes by maximizing B-cell depletion and optimizing disease control.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (approved 600 mg dose of ocrelizumab) **Blinding:** Double-blind **Randomization:** Randomized (Parallel Group Assignment)

3.2 Planned Enrollment & Duration Target N: 864 participants **Number of Sites:** 123 locations globally **Estimated Study Start:** 26-Nov-2020 **Estimated Primary Completion:** 19-Dec-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 55 years old with a diagnosis of Relapsing Multiple Sclerosis (RMS). 2. At least two documented clinical relapses within the last 2 years prior to screening, or one clinical relapse in the year prior to screening. No relapse 30 days prior to screening and at baseline. 3. Participants must be neurologically stable for at least 30 days prior to randomization and baseline. 4. Expanded Disability Status Scale (EDSS) score at screening and baseline from 0 to 5.5 inclusive. 5. Average Timed 25-Foot Walk Test (T25FWT) score over two trials at screening and over two trials at baseline respectively, up to 150 (inclusive) seconds. 6. Average 9-Hole Peg Test (9-HPT) score over four trials at screening and over four trials at baseline respectively, up to 250 (inclusive) seconds. 7. Documented magnetic resonance imaging (MRI) of the brain with abnormalities consistent with MS at screening. 8. Participants requiring symptomatic treatment for MS and/or physiotherapy must be treated at a stable dose. No initiation of symptomatic treatment or physiotherapy within 4 weeks of randomization. 9. Females of childbearing potential must agree to remain abstinent or use adequate contraceptive methods. Female participants without reproductive potential may be enrolled (e.g., if post-menopausal or surgically sterile).

4.2 Exclusion Criteria 1. History of primary progressive MS (PPMS) at screening. 2. Any known or suspected active infection at screening or baseline (except nailbed infections), or any major episode of infection requiring hospitalization or treatment with IV antimicrobials within 8 weeks or treatment with oral antimicrobials within 2 weeks, prior to and during screening. 3. History of confirmed or suspected progressive multifocal leukoencephalopathy (PML). 4. History of cancer, including hematologic malignancy and solid tumors, within 10 years of screening. 5. Immunocompromised state or history of currently active primary or secondary, non-drug-related, immunodeficiency. 6. Receipt of a live or live-attenuated vaccine within 6 weeks prior to randomization. 7. Inability to complete an MRI or contraindication to gadolinium administration. 8. Contraindications to mandatory pre-medications for infusion-related reactions (IRRs). 9. Known presence of other neurologic disorders that could interfere with the diagnosis of MS or assessments. 10. Any concomitant disease that may require chronic treatment with systemic corticosteroids or immunosuppressants. 11. Significant, uncontrolled disease that may preclude participation. 12. Pregnant or breastfeeding or intending to become pregnant. 13. Lack of peripheral venous access. 14. History of alcohol or other drug abuse within 12 months prior to screening. 15. Treatment with any investigational agent within 24 weeks prior to screening or any experimental procedure for MS. 16. Previous use of anti-CD20s (including ocrelizumab), unless the last infusion was more than 2 years before screening, B-cell

count is normal, and treatment stop was not motivated by safety reasons or lack of efficacy. 17. Previous treatment with fingolimod, siponimod, or ozanimod within 6 weeks of baseline; natalizumab within 4.5 months; or interferons beta / glatiramer acetate within 2 weeks. 18. Any previous treatment with mitoxantrone, cladribine, atacicept, alemtuzumab, and daclizumab, or any other immunomodulatory/immunosuppressive medication without appropriate washout. 19. Any previous treatment with bone marrow transplantation, hematopoietic stem cell transplantation, or history of transplantation/anti-rejection therapy. 20. Treatment with IV immunoglobulin (Ig) or plasmapheresis within 12 weeks prior to randomization. 21. Systemic corticosteroid therapy within 4 weeks prior to screening. 22. Positive screening tests for active, latent, or inadequately treated hepatitis B. 23. Sensitivity or intolerance to any ingredient (including excipients) of ocrelizumab.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * *Experimental Group (Higher Dose):* 1200 mg (participant body weight < 75 kg) or 1800 mg (participant body weight $\geq$ 75 kg) administered by IV infusion every 24 weeks (Q24W). * *Active Comparator Group:* 600 mg administered by IV infusion every 24 weeks (Q24W). **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** Minimum of 120 weeks in the double-blind treatment (DBT) phase. Eligible participants may transition to an optional higher dose open-label extension (OLE) phase for approximately 96 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Mandatory premedication consisting of methylprednisolone (or equivalent) and an antihistaminic drug (e.g., chlorcyclizine) administered prior to each ocrelizumab infusion to manage infusion-related reactions (IRRs). **Prohibited:** Live or live-attenuated vaccines within 6 weeks prior to randomization. Treatment with any other investigational agent or immunosuppressant/immunomodulatory therapy without an appropriate washout period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -42 to 0 Informed Consent, Medical History, MRI, EDSS, Infection screening Baseline
Day 0	Randomization, First IV Infusion (split into 2 half-doses, 2 weeks apart), PK Labs	Interim	Every 24 Weeks
Q24W Maintenance Infusions, Efficacy Assessments (EDSS, T25FWT, 9-HPT), Safety Labs	End of DBT	Week 120 Final Double-Blind Efficacy Assessments, Optional OLE Entry, IP Accountability	

(Note: Participants not entering or completing the OLE proceed to a 48-week safety follow-up and subsequent B-cell monitoring (BCM) phase until B-cell levels replete to baseline or the low level of normal [LLN])

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Time to Onset of 12-week Composite Confirmed Disability Progression (cCDP12): Time to onset of cCDP is defined as the first occurrence of a predefined confirmed progression event measured by Expanded Disability Status Scale (EDSS), Timed 25-foot Walk Test (T25FWT) or 9-hole Peg Test (9-HPT).

7.2 Secondary & Exploratory Endpoints * Time to Onset of 24-week cCDP (cCDP24). * Time to Onset of 48-week cCDP (cCDP48). * Time to Onset of cCDP12 Independent of Protocol-defined Relapses (PDR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation throughout the trial focusing on infusion-related reactions (IRRs) and respiratory/opportunistic infections. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timelines to the sponsor and regulatory authorities for serious and unexpected AEs. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) utilized to periodically review unblinded safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy analysis; Safety Set for all participants who received at least one dose of the study drug. **Sample Size Justification:** Approximately 864 participants randomized to provide sufficient statistical power ($\alpha = 0.05$) to detect a meaningful delay in the time to 12-week cCDP. **Handling of Missing Data:** Time-to-event endpoints analyzed using survival methods (e.g., Kaplan-Meier estimates, stratified Cox proportional hazards regression) incorporating censoring for participants lost to follow-up or completing the study without an event.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite monitoring and centralized data

review to assure protocol compliance. **Audit Trail:** Utilizing 21 CFR Part 11 compliant Electronic Data Capture (EDC) systems with eCRF locking and standardized data clarification forms.

11. Study Identifiers & Governance

Primary ID: BN42082 **Registry Number:** NCT04544436 **Posting ID:** 909857978771445983 **Sponsor/Lead Organization:** Roche **Study Title:** MUSETTE **Official Regulatory Title:** A Phase IIIb Multicenter, Randomized, Double-blind, Controlled Study to Evaluate the Efficacy, Safety and Pharmacokinetics of a Higher Dose of Ocrelizumab in Adults With Relapsing Multiple Sclerosis

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Relapsing Multiple Sclerosis (RMS) **Diagnosis Threshold:** Diagnosis of RMS with ≥ 2 clinical relapses within 2 years or 1 relapse within 1 year; EDSS score 0 to 5.5 **Medical Methodology:** Disease-Modifying Therapy (Anti-CD20+ B-cell depletion therapy) **Optimization Target:** Delay or prevention of confirmed disability progression (cCDP)

13. Study Procedures & Methodology

Management Pathway: Q24W Intravenous (IV) Infusion **Education Protocol (HCP):** Pre-medication protocol administration and Infusion-Related Reaction (IRR) management training **Education Protocol (Patient):** Contraception requirement counseling, infection risk awareness, and reporting of emerging MS symptoms **Quality Audit Mechanism:** Centralized independent review for MRI assessments and standardized rater training for EDSS, T25FWT, and 9-HPT clinical tests

14. Observation & Follow-Up Schedule

Total Duration: Minimum 120 weeks (DBT phase) + ~96 weeks (OLE phase) + 48 weeks safety follow-up + B-cell monitoring (BCM) phase **Onsite Visit Cadence:** Every 24 weeks (Q24W) for infusions and major clinical efficacy evaluations **Remote Monitoring Frequency:** As per institutional standards and interim safety check-ins **Checklist Review Interval:** Every 12-24 weeks for key disability progression metrics (EDSS, T25FWT, 9-HPT)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |